

SEVERE-WEATHER EVIDENCE REVIEW

Hail near an Okeene-area property

6 May 2024 · 18:00–19:30 CDT · exact street address and claim identifiers withheld

EVIDENCE SUPPORTS HAIL IN THE IMMEDIATE AREA

Finding

Radar-derived estimates and nearby ground reports place hail in the immediate property area during the stated loss window. The closest Local Storm Report was 0.31 miles away at 18:35 CDT; it recorded 1.25-inch hail. A second report 0.46 miles away recorded 1.75-inch hail ten minutes later.

NOAA's archived 60-minute MESH grid at 18:40 CDT estimated 1.67 inches at the nearest grid cell and 1.70 inches within 0.6 miles. This supports nearby hail occurrence; it does not establish impact or damage at the structure.

CASE REFERENCE

CW-OK-20240506

HAZARD

Hail

REVIEW RADIUS

25 miles

ARCHIVE CHECK

17 Sep 2026

REPORT STATUS

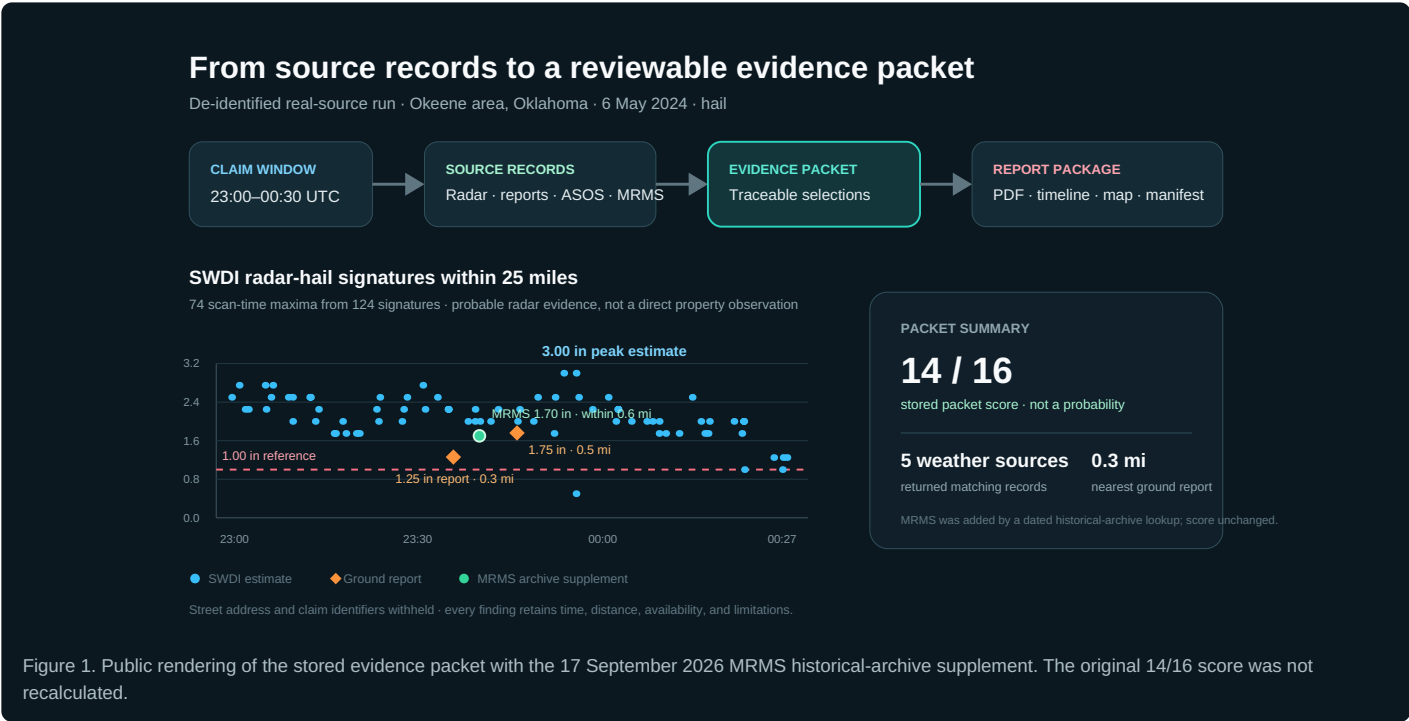
Public sample

0.31 mi	1.70 in	124	14 / 16
NEAREST GROUND REPORT	MRMS MAXIMUM WITHIN 0.6 MI	SWDI HAIL SIGNATURES	STORED EVIDENCE SCORE

SOURCE ACCOUNTING

WEATHER SOURCE	RECORDS USED	STATUS	ROLE
NOAA MRMS MESH	Historical 60-minute grid	Matched	Property-area radar estimate
NCEI SWDI nx3hail	124 signatures / 74 scan times	Matched	Radar-hail timing and area magnitude
IEM Local Storm Reports	4 hail reports	Matched	Time-, size-, and distance-qualified reports
NCEI Storm Events	6 hail records	Matched	Delayed historical report archive
IEM ASOS	4 stations	Matched	Nearby convective-weather context

Five weather-data sources returned matching records. Property geocoding is a routing input and is not counted as weather evidence. Local Storm Reports and Storm Events can represent the same underlying observations and are not assumed to be independent.



EVENT SEQUENCE

18:00	Claim window opens.
18:10	2.75-inch hail report, 16.44 miles WSW.
18:35	1.25-inch hail report, 0.31 miles NW.
18:40	MRMS MESH: 1.67 inches at nearest grid cell; 1.70 inches within 0.6 miles.
18:45	1.75-inch hail report, 0.46 miles NE.
18:57	1.50-inch hail report, 12.00 miles WSW.
19:30	Claim window closes.

WHAT EACH RECORD CONTRIBUTES

MRMS MESH

Gridded, multi-radar estimate near the property. It is an estimate of expected hail size, not a surface measurement.

SWDI nx3hail

NEXRAD Level-III hail signatures establish timing and radar-indicated hail across the search area. The 3.00-inch value is the area peak, not a property reading.

Local Storm Reports

Two reports under 0.5 miles provide close ground-report context. Their coordinates and magnitudes still carry reporting and location uncertainty.

ASOS observations

Nearby stations record convective conditions, including thunderstorm and heavy-rain signals. They do not measure hail at the property.

REASONED INTERPRETATION

The strongest evidence is the convergence of a property-area MRMS estimate and two time-matched ground reports within half a mile. SWDI adds repeated radar-hail signatures through the window. Together these records support hail near the property; none directly observes roof impact, damage, or causation.

STORED FACTOR SCORE

Radar beam quality	3 / 3	Radar magnitude	3 / 3
Temporal persistence	2 / 2	Ground-report proximity	4 / 4
Radar/report size agreement	0 / 2	Station corroboration	2 / 2

Deterministic evidence score

14 / 16

This is the score stored by the original packet. It summarizes configured evidence factors; it is not a probability. MRMS was added later and the score was deliberately left unchanged.

MRMS ARCHIVE CHECK

The review queried NOAA's public `noaa-mrms-pds` archive for `MESH_Max_60min_00.50` at selected times in the event window. The 23:40 UTC file returned 42.3 mm at the nearest grid cell and 43.1 mm within 0.6 miles. Values were converted with 25.4 mm per inch.

The earlier no-match was caused by querying the recent NCEP directory for a historical event. Archive presence and property-area values were verified directly; the full claim workflow was not rerun.

LIMITS ON USE

- MESH and SWDI values are radar-derived estimates. They are not physical hail measurements at the structure.
- A report's plotted point can differ from where hail fell. Distances are to reported coordinates.
- IEM Local Storm Reports and NCEI Storm Events can contain related records; counts must not be read as independent confirmations.
- ASOS stations are 18–44 miles from the property and provide regional context only.
- The 3.00-inch SWDI maximum occurred within the 25-mile search area, not necessarily at the property.
- This review addresses meteorological evidence. It does not determine damage, repair scope, policy coverage, or claim liability.

Privacy and reproducibility. The public sample removes the street address, exact property coordinates, claim identifiers, raw report narratives, and source query bounding boxes. Times, rounded distances, magnitudes, source names, and method notes are retained so the reasoning remains reviewable.

SOURCE DOCUMENTATION

1. [NOAA Multi-Radar/Multi-Sensor System on AWS](#) — public historical archive.
2. [NSSL operational MRMS tables](#) — MESH product cadence, units, and product definitions.
3. [NCEI Severe Weather Data Inventory](#) — SWDI scope and limitations.
4. [NCEI severe-weather products](#) — Storm Events and radar-data access.
5. [Iowa Environmental Mesonet LSR archive](#) — archive schema and completeness notice.

Weather records are attributed to their source organizations. No endorsement or affiliation is implied. Review prepared 17 September 2026.